

3.2.3 Activity: Arithmetic Calculator Using Functions

کیلکولیٹر میں ہر عمل الگ فنکشن ہے: جمع، تفریق، ضرب، تقسیم۔

CS XII

Chapter 3: Programming Fundamentals

Lecture 12 of 17

3.2.3 Activity: Arithmetic Calculator Using Functions

LEARN IT

Each arithmetic operation is a **separate function**. That is modularity: addition is one small task, division is another (with a guard against dividing by zero). The main program only asks for two numbers and calls the functions.

```
def add(a, b):
    return a + b

def subtract(a, b):
    return a - b

def multiply(a, b):
    return a * b

def divide(a, b):
    if b != 0:
        return a / b
    else:
        return "Cannot divide by zero"

num1 = float(input("Enter first number: "))
num2 = float(input("Enter second number: "))
print("Addition:", add(num1, num2))
print("Subtraction:", subtract(num1, num2))
print("Multiplication:", multiply(num1, num2))
print("Division:", divide(num1, num2))
```

WORKED EXAMPLE

Sample run with 12 and 4. `float` is used so decimals work; even whole numbers print with a decimal (12 becomes 12.0).

OUTPUT

```
Enter first number: 12
Enter second number: 4
Addition: 16.0
Subtraction: 8.0
Multiplication: 48.0
Division: 3.0
```

Check: $12 + 4 = 16$, $12 - 4 = 8$, $12 \times 4 = 48$, $12 \div 4 = 3$. In Python 3, `/` always produces a float, so 3.0 not 3.

Zero divisor:

```
def divide(a, b):  
    if b != 0:  
        return a / b  
    else:  
        return "Cannot divide by zero"  
  
print(divide(10, 0))  
print(divide(10, 2))
```

OUTPUT

```
Cannot divide by zero  
5.0
```

Without the `if b != 0` test, `10 / 0` would raise `ZeroDivisionError` and stop the program.

REMEMBER IT

Four functions, four names: `add`, `subtract`, `multiply`, `divide`. The main program should look like a contents page — a sequence of calls — not a pile of formulae.

COMMON MISTAKE

`input()` returns a string. If you skip `float()` (or `int()`), `add` would concatenate: `"12" + "4"` is `"124"`. Convert first. Also, `divide` returns either a number or a string; do not later do arithmetic on the error message.

اردو میں سمجھیں

کیلکولیٹر میں ہر عمل الگ فنکشن ہے: جمع، تفریق، ضرب، تقسیم۔ یہ ماڈیولریٹی ہے۔ تقسیم سے پہلے صفر کی جانچ ضروری ہے ورنہ پروگرام رک جاتا ہے۔ `input()` سے ملنے والی تار کو `float()` سے عدد بنانا لازمی ہے ورنہ جمع جوڑنے کے بجائے الفاظ جوڑ دے گی۔

CHECK YOURSELF

1. What does `divide(10, 0)` return in this activity?
2. Why are the numbers converted with `float()` ?

Answers: 1. The string "Cannot divide by zero". 2. Because `input()` returns text; `float()` turns it into a number so `+` `-` `*` `/` do arithmetic, including decimals.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. A calculator activity should use:

- a. A function per operation
- b. One 200-line print spaghetti only
- c. No functions
- d. Only globals named a,b,c,d,e

MCQ 2. Division must consider:

- a. Zero divisor
- b. NAND grouping
- c. OSI layer
- d. Figma frames

MCQ 3. Menu + functions is:

- a. Selection + modularity
- b. A stack hardware
- c. A pie
- d. A foreign key

MCQ 4. return lets you:

- a. Print nicely in one place
- b. Hide errors
- c. Skip tests
- d. Avoid operators

MCQ 5. float(input) may raise:

- a. ValueError
- b. A pie
- c. A K-map
- d. A sprint

MCQ 6. This activity prepares:

- a. Reusable functions + later try/except
- b. Only Access
- c. Only HCI colours
- d. Only ER

Extra short questions

1. Outline four functions for $+$ $-$ $\times$ $\div$.
2. How do you stop divide by zero?
3. Where does the menu live?
4. Why not copy the formula 12 times?

Extra long question

1. Write a mini calculator with four functions, a loop menu, and a safe divide. Sample session of 3 operations then quit.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. add, sub, mul, div with two params, return.
2. if b==0: return a message / raise / skip.
3. In main: while True: choice...
4. One fix in one function.

Long

1. Working structure, sample I/O, ZeroDivision handled. Mention return vs print.